

Prime Factorization Visualization without Tensors

August 2025

A prime factorization can be visualized as a square or a cube.

5×7 is a square of 5 rows and 7 columns

$5 \times 7 \times 2$ is a cube of 5 rows, 7 columns, and 2 slabs

This breaks down when our factorization has more than 3 factors, since tend to visualize in at most 3 dimensions.

To visualize $5 \times 7 \times 2 \times 3$ we can use the source representation shown on the next page.

